MATH7005 2026/2027
Assignment and Project Specification

5 October 2026 (revised 28 October 2026)

1 Key information

- **Initial release date:** 5 October 2026.
- **Assignment deadline:** 26 October 2026.
- **Group registration deadline:** 6 November 2026. Email the lecturer your group members' names and your chosen project idea. No changes to group formation are allowed after this date.
- **Project submission deadline:** 16 November 2026.
- **Assessment week:** 17–22 November 2026, for the group demo day and the individual AI-assisted question-and-answer sessions.
- The **assignment is an individual assessment**. The **project is a group assessment** for groups of one to three students (see Section 5.1). The question-and-answer session remains **individual**.
- The assignment accounts for **10%** and the project accounts for **90%** of the subject's final mark.

Revision note (28 October 2026)

This revision incorporates the announcement of 25 October 2026. The project is now completed in groups of one to three students; the project deadline is 16 November 2026; and all project assessment takes place during 17–22 November 2026. The demo day is a demonstration of the working product only, and the question-and-answer session is held individually in a computer lab. Sections 4, 5, and 6 have been updated accordingly.

2 Introduction

In MATH7005, you have learned a range of essential data analytics and machine learning techniques, together with substantial Python programming. In particular, you have developed practical experience with NumPy and pandas for data manipulation, analysis, and computation. These skills are highly useful in both academic and business settings. The assignment and project give you an opportunity to apply them to a real-world problem by proposing and developing a data product.

Entrepreneurial experience is highly valued in today's job market. You are encouraged to consider how your work could become a viable venture.

The assessment is designed to strengthen your data analytics, machine learning, product-design, coding, communication, and critical-thinking skills. Each project group must design and build at least part of a data analytics or machine learning tool centred on the annual theme.

3 Theme

For the 2026–2027 assessment, the theme is:

A Python-powered data product using Hong Kong open data to improve life, services, or economic opportunities in Hong Kong.

The theme is deliberately broad enough to support diverse ideas while retaining a clear focus on Hong Kong and the practical use of local open data. Projects may address areas such as transport, public services, health, the environment, housing, education, culture, tourism, commerce, or intellectual property. The proposed product must address a clearly defined problem and provide value to a specific group of users.

4 Assignment: video pitch (10%)

Each student must submit a three-minute video pitch consisting of voice-over slides. The pitch should establish the context for the data product that the student proposes to develop for the project.

The pitch should concisely explain:

- the problem or opportunity being addressed;
- the intended users and their needs;
- the proposed data product and its value;
- the relevant data that may be used; and
- the intended social, economic, or practical impact.

This lightly weighted task is intended to keep students on track and to support the early development of a feasible project. The deadline is **26 October 2026**. Submission instructions will be provided on Moodle.

The assignment is marked **individually**, even if you later work with other students on the project. Identical assignment submissions are not permitted: if two or more students submit identical assignments, each of them will receive **zero for the assignment**.

5 Project (90%)

5.1 Group formation

The world is increasingly moving towards a "one-person company" (OPC) style, and it is becoming rarer for several people to start working on the same project from the very beginning. For this reason, the assessment is deliberately arranged in two stages: each student first develops their own idea in the individual assignment, and then takes that idea to find like-minded partners for the project. The communication and persuasion this requires are themselves skills that matter in the AI era.

- A project group has **one to three members**. A group of one, in which a student completes the whole project alone, is allowed.
- Smaller groups are **not at a disadvantage**. The expected scope and effort of the project are adjusted according to the number of members in the group.
- By **6 November 2026**, each group must email the lecturer the names of all members and the chosen project idea. Students working alone must also send this information. **No changes to group formation are allowed after 6 November 2026.**
- The group's project idea may be developed from one member's assignment pitch or from a combination of members' pitches.

5.2 Problem setting and required work

Imagine that you are startup founders building an innovative, publicly usable data-mining tool. The tool may take the form of a website, mobile application, desktop application, smart device, or another suitable product. Your work should include, but is not limited to, the following:

1. Design a data product for the theme. Address fundamental questions such as:
   - Who are the intended users?
   - Is the product goal realistic?
   - What makes the product unique?
   - What impact could the product have on society?
2. Select suitable datasets. Every project **must use one or more datasets from the Hong Kong Government's open-data portal, https://data.gov.hk/en/, as a main part of its data and analysis**. Supplementary data may also be obtained from sources such as:
   - datasets from Kaggle, such as https://www.kaggle.com/;
   - intellectual property datasets; and
   - other sources relevant to the theme.
3. If the available data are insufficient, propose additional "good-to-have" datasets. Specify the key attributes or fields, explain why they are important, and identify who may be able to provide the data. You may use Python to generate realistic synthetic data where appropriate, but you must clearly label and justify such data.
4. Propose and build at least a partial product prototype. A fully functional product is not required, but the prototype must contain working features.
5. Apply relevant modelling and programming techniques to the downloaded and/or generated data, and explain how each technique advances the product goals. All submitted code must be written in Python.
6. **For students who wish to achieve an outstanding mark:** Research the machine learning literature and select an original research paper published in or after 2022 that has more than 2,022 citations. Survey and review papers do not count. Discuss how the selected paper's techniques could be applied to your data product.
7. Use suitable data-mining, data-science, machine-learning, or AI methods from this subject or beyond it. However, do not base the project primarily on techniques covered in other ORBS subjects. For example, a project primarily based on ARIMA and the elbow method for PCA will receive a very low score.
8. Propose future steps needed to bring the data product to life.

5.3 Project deliverables and assessment

Component	Done as	When
Final report and code	Group (one submission)	By 16 November 2026
Live product demonstration (demo day)	Group	17–22 November 2026
AI-assisted question-and-answer session	Individual	17–22 November 2026

1. **Final report and code (group)**

   Each group submits **one** consolidated report and **one** set of code, documenting the steps taken to achieve the group's goals. Group members do not submit separate reports. Extensive background material is unnecessary unless it primarily reflects the group's own contribution.

   - There is no word limit, but the report should comprehensively record what the group did.

- Include the complete code as an attachment or in the format specified on Moodle.
- **Groups of two or three** must include a **contribution statement** setting out the division of work and each member's individual contributions, including the parts of the product, analysis, and code that each member was primarily responsible for. Clearly indicate in the code (for example, in module or function docstrings) who wrote each main component. A group of one does not need a contribution statement.
- You may describe failed attempts when they demonstrate effort, reflection, or learning.
- No resubmission is allowed after the deadline.
- The report and associated materials must be submitted on Moodle by **16 November 2026**.

2. **Live product demonstration (group demo day)**

   Each group gives a **live demonstration** of its finished data product during the assessment week. The demo day is **purely a demonstration of the final working product**. Do not present background, motivation, data description, or methodology, as these are already covered in the final report. **Slides are not encouraged**, unless they are a vital part of the demonstration. Members of a group present together and may share the demonstration between them. The time allocated to each group, and the schedule, will be announced on Moodle.

3. **Question-and-answer session (individual)**

   Each student attends an **individual AI-assisted question-and-answer session**, of approximately **30 minutes**, held in a **computer lab** during 17–22 November 2026. As part of the assessment process, AI will collaborate with the human assessor to analyse the submitted report and code; the questions are drawn from these submissions, and the human assessor compiles the final questions.

   This session is **strictly individual**, even for students in a group. **No communication between group partners, or with anyone else, is permitted during the session.**

   Questions may cover **any component of the project**, not only the parts a student is credited with in the contribution statement. Every student must therefore understand every component of the group's project and be able to explain and defend it, including the design choices, modelling, implementation, code, results, limitations, and use of AI. The schedule will be announced on Moodle.

5.4 Use of AI

AI assistance is allowed for the project, provided that it is clearly declared. The final report must identify the AI tools used, describe how they were used, and, in a group, state which members used them. Students remain responsible for the correctness, originality, and integrity of all submitted work, regardless of whether it was written by a group member or generated with the assistance of AI.

6 Grading

The project will be graded broadly on the following aspects. Expectations are adjusted according to group size. Members of the same group will not necessarily receive the same mark: each student's project mark reflects the group's deliverables, that student's contributions, and their performance in the individual question-and-answer session.

1. **Understanding of subject knowledge.** Demonstrate a broad set of relevant skills by applying multiple data-mining tasks to the proposed data product. Do not limit the work to only one or two methods.

2. **Understanding of Python coding.** Demonstrate a thorough understanding of all submitted Python code, including code written by any group member and code generated or modified with AI assistance.

   - Code should be well documented and easy to understand.
   - Main Python functions should meaningfully implement the algorithms. One-line calls to scikit-learn functions will not be accepted as the primary implementation.

3. **Uniqueness and innovation.** Explain why the data product is distinctive, innovative, and useful. The "productness" of the work is important: do not submit something that is merely a report and cannot be used by its intended audience.

4. **Report quality.** Present the problem, data, methods, product, findings, limitations, AI use, and future work clearly and rigorously.

5. **Live demonstration and responses.** Demonstrate a functioning product as a group, and answer questions in the individual session accurately, critically, and independently.

6. **Originality and academic integrity.** Every group is expected to submit a distinct solution. Members of the same group work together on a shared product, but discussion with students outside your group should not extend to sharing work: submitting substantially similar products across groups is not acceptable and may result in reduced marks or further action under the University's academic integrity procedures. The individual assignment must be each student's own work; identical assignment submissions receive zero.